

■ Bubbly – Macro Bubble Detection Framework

1. Vision and Philosophy

Bubbly is a research-grade framework designed to quantify and visualize market bubble dynamics in real time. Its core principle is: "Everyone can say it's a bubble — the hard part is timing it." Bubbly aims to create a quantitative early warning system for financial instability by blending valuation metrics, macro liquidity indicators, and behavioral signals (like volatility complacency). It maps the market's psychological and macroeconomic state into an interpretable "bubble pressure index".

2. Objectives

1. Aggregate multiple macro & market indicators into a unified index. 2. Normalize each variable statistically (z-scores). 3. Reconstruct past bubbles (Dot-Com, 2008, Crypto 2021) for validation. 4. Automate weekly reporting with HTML/PDF outputs. 5. Evolve toward adaptive, data-driven thresholds and regime-switching models.

3. Core Indicators

• Buffett Ratio (Market Cap / GDP): valuation pressure indicator, peaked before 2000 and 2021 bubbles. • Shiller CAPE: cyclically adjusted P/E ratio reflecting long-term optimism. • M2 YoY Growth: liquidity driver; high in 2020, negative before 2022 crash. • VIX: volatility index capturing investor complacency and fear.

4. Composite Bubble Pressure Index

Weighted z-score model: $B = 0.4 * Z(\text{Buffett}) + 0.4 * Z(\text{CAPE}) - 0.1 * Z(\text{M2}) - 0.1 * Z(\text{VIX})$

Interpretation: • Positive B → overheating/speculative pressure. • Negative B → undervaluation or contraction. Thresholds: - $B < 1.0$ → Expansion (healthy) - $1.0 \leq B < 2.0$ → Euphoria (overheated) - $B \geq 2.0$ → Instability (bubble risk)

5. Indicators and Roles

Valuation: Buffett Ratio, CAPE Liquidity: M2, Fed Funds Rate, Balance Sheet Behavioral: VIX, Margin Debt, Put/Call Ratio Macro backdrop: GDP growth, Inflation, Real Yields Future additions: real rates, margin debt, profit margins, crypto correlations.

6. Historical Validation

Reconstruct past bubbles with the same model: 2000 Dot-Com → Extreme CAPE, Buffett ratio
2007 Credit Bubble → Moderate valuation, loose liquidity 2021 Everything Bubble → High CAPE & Buffett, negative M2 later 2009 Low → Negative B, start of secular bull Goal: correlate B with future market drawdowns and crisis events.

7. Analytical Extensions

- Regime labeling: Expansion, Euphoria, Instability, Correction.
- ML: Markov or Bayesian regime switching models.
- Adaptive thresholds: percentile-based dynamic cutoffs.
- Visualization: Plotly dashboard with B history and weights.

8. System Implementation

Pipeline: 1. Fetch data (yfinance, Shiller .xls, optional FRED). 2. Compute normalized z-scores. 3. Combine into weighted index. 4. Output HTML reports (bubbly_report.html). Example Output: Buffett 2.71 | CAPE 1.88 | M2 -0.54 | VIX -0.40 → Composite 0.91 → Expansion.

9. Long-Term Vision

- VPS automation with daily updates.
- Add sentiment, yield spreads, crypto metrics.
- Use AI for cross-market contagion detection.
- Replay 1990–present with rolling B simulation.

10. Missing Dimensions

Potentially overlooked metrics: Credit/leverage → Margin debt, corporate spreads Housing → Case-Shiller ratio Real yields → 10y – CPI Liquidity proxies → Reverse repo, Fed balance sheet Global risk → USD index, EM flows Sentiment → AAI, Fear-Greed index

11. Validation Goals

Steps: 1. Collect historical monthly data since 1980. 2. Compute B through time. 3. Compare to >15% SP500 drawdowns. 4. Measure hit rates, false positives, and signal lead time.

12. Conclusion

Bubbly is evolving into an interpretable early-warning framework for systemic market risk. It merges economics, behavioral finance, and data science into a transparent structure. With calibration and expansion, it can become a valuable open-source research tool for identifying speculative regimes and financial bubbles in real time.